

MHF 4U EXAM PREP QUESTIONS

This package contains some but not all of the topics covered in this course. It should be used as supplemental material to reviewing your notes, tests, quizzes and review exercises from your textbook.

POLYNOMIAL FUNCTIONS

1. Divide and state restrictions on the variables.

- a) $(x^2 + 10x + 16) \div (x + 2)$ b) $(y^3 + y^2 - 2) \div (y - 1)$
c) $(3y^3 - 2y^2 + 12y - 9) \div (y^2 + 2)$ d) $(15x - 4x^3 - 9x^2 + 3x^4 - 4) \div (3x - 4)$

2. Determine the remainder without dividing.

- a) $(x^2 + 5x - 8) \div (x - 2)$ b) $(3m^2 + 7m + 1) \div (m + 3)$
c) $(2x^2 + 5x + 11) \div (2x - 1)$ d) $(8y^3 + 12y^2 - 4y + 5) \div (2y + 3)$

3. Find the value of k if the remainder is -1 when $(x^3 - 3x^2 - 6x + k)$ is divided by $(x + 2)$

4. Grace has recently written a book. Her publisher models the cumulative total number of copies sold, $C(t)$, t days after the book is released by the formula, $C(t) = 0.5t^2 + 20t + 50$; $t \in [0, 30]$. Divide $C(t)$ by $t - 10$ and state the quotient and the remainder. What does the quotient represent in this case?

5. Find the integer k such that $(x - 2)$ divides $(3x^3 - 2x^2 + x + k)$ evenly.

6. Factor completely.

- a) $(x^3 - x^2 - 5x - 3)$ b) $(x^3 + 5x^2 + 3x - 4)$
c) $(2x^3 - x^2 - 2x + 1)$ d) $(3x^3 + 13x^2 - 16)$

7. Solve and check.

- a) $y^3 = 9y$ b) $n^3 - 3n - 2 = 0$
c) $3w^2 + 11w = 2w^3 + 6$ d) $3x^2 - 2 = 8x^3 - 7x$

8. A cereal box is a rectangular prism with a volume of 2500cm^3 . The box is 4 times as wide as it is deep, and 5cm taller than it is wide. What are the dimensions of the box?

9. Graph each function and determine all key properties (domain, range, intercepts, intervals where $f(x) > 0$, $f(x) \leq 0$, end behaviour).

- a) $f(x) = x(x + 4)(x - 4)$ b) $f(x) = -(x + 2)(x - 3)(x - 5)$
c) $f(x) = -x(x - 3)(x + 1)(x + 4)$ d) $f(x) = x^2(x^2 - 9)$
e) $f(x) = x^4 - 2$ f) $f(x) = x^3 - x^2 - 2x - 3$

10. Solve each inequality.

a) $x(x+3) > 0$

b) $(x+4)(x-1) < 0$

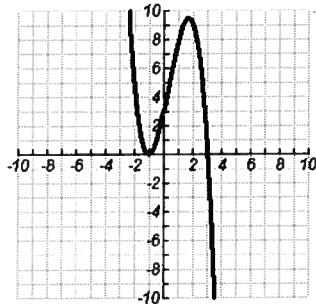
c) $(x+2)(x-2)(x+4) \geq 0$

d) $x^3 - 2x^2 - 5x > 6$

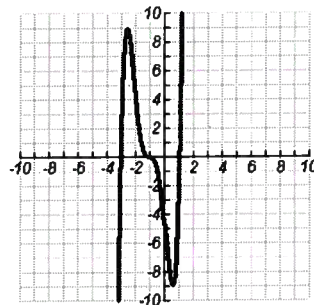
11. A quartic function has x-intercepts of -5, -1, 2, and 4. The point (3, -8) lies on the curve. Determine the equation of this function.

12. Determine the equation of each function graphed below.

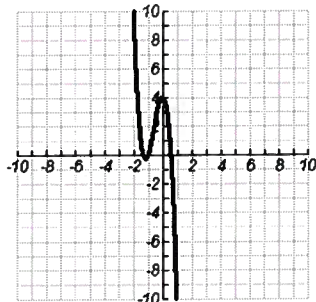
a)



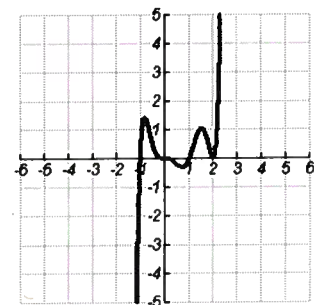
b)



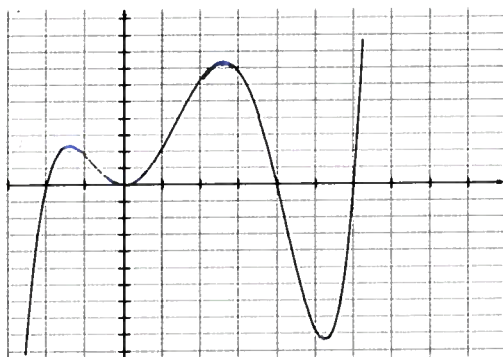
c)



d)



13. Given:



- Is the degree of the function even or odd?
- Is the leading coefficient positive or negative?
- State the number of zeroes and their value

ANSWERS: POLYNOMIAL FUNCTIONS

- 1a) $x+8, x \neq -2$ b) $y^2+2y+2, y \neq 1$ c) $3y-2, R: 6y-5$ d) $x^3-3x+1, x \neq 4/3$
 2a) 6 b) 7 c) 14 d) 11 3) 7 4) $Q(t)=0.5t+25, R(t)=300$ 5) $k=-18$
 6a) $(x+1)^2(x-3)$ b) $(x+4)(x^2+x-1)$ c) $(x-1)(x+1)(2x-1)$ d) $(x-1)(x+4)(3x+4)$
 7a) $0, \pm 3$ b) $-1, 2$ c) $\frac{1}{2}, 3, -2$ d) $1, \frac{-5 \pm \sqrt{65}}{10}$ 8) $25\text{cm} \times 20\text{cm} \times 5\text{cm}$
 10a) $x \in (-\infty, -3) \cup (0, \infty)$ b) $x \in (-4, 1)$ c) $x \in [-4, -2] \cup [2, \infty)$ d) $x \in (3.76, \infty)$
 11) $y = \frac{1}{4}(x+5)(x+1)(x-2)(x-4)$
 12a) $y = -(x+1)^2(x-3)$ b) $y = -2(x+1)^3(x+3)(x-1)$
 c) $y = a(2x+3)(x+1)(2x-1)$ d) $y = ax^3(x-2)^2(x+1)(x-1)$
 13 a) odd b) positive c) 2, 0, 4, 6 double at 0

RATIONAL FUNCTIONS

1. Determine the equation(s) of all vertical asymptotes and sketch the curve near the asymptotes.

- a) $f(x) = \frac{4}{x-2}$ b) $f(x) = \frac{5}{(x-3)^2}$ c) $f(x) = \frac{x}{1-x^2}$
 d) $f(x) = \frac{4x^4+8x}{x^2+6x+8}$ e) $f(x) = \frac{x+3}{x^4-9x^2}$ f) $f(x) = \frac{3}{x^3-4x}$

2. Determine the equation(s) of all vertical, horizontal or slant asymptotes and use these to sketch each curve.

- a) $f(x) = \frac{2x-3}{5-x}$ b) $f(x) = \frac{x^2+4}{x}$ c) $f(x) = \frac{9x-6}{1-3x}$
 d) $f(x) = \frac{x^2+5x+4}{x-1}$ e) $f(x) = \frac{x^3+5x^2+3x+10}{x^2+2}$ f) $f(x) = \frac{6x^2+4x+1}{5-3x^2}$

3. Given $y = \frac{x^2-25}{x-5}$ (a) Determine the equation of any vertical and horizontal asymptotes and the corresponding function behaviour. (b) domain
 (c) Determine the x-intercept. (d) Graph

4. Given $h(x) = \frac{x^2+6x+5}{(x+5)^2(x-1)}$ (a) Determine the equation of the vertical and horizontal asymptotes and the corresponding function behaviour. (b) domain
 (c) Determine the x-intercept. (d) Graph

5. Given $f(x) = \frac{x^2-x-6}{x-2}$ (a) Determine the equation of the vertical and oblique(slant) asymptotes and the corresponding function behaviour. (b) domain
 (c) Determine the x-intercept. (d) Graph

6. A piece of machinery depreciates in value, V , in dollars, over time, t , in months. The value is given by the function, $V(t) = 5000 - \frac{2000t^2}{(t+2)^2}$.

a) Determine the values of the machinery after i) 6 months, and ii) 10 years.

- b) Describe the end behaviour of this function (what happens to the value of the machinery in the long run)?
 c) Will the machinery ever have a value of \$0? Explain.

ANSWERS: RATIONAL FUNCTIONS

- 1a) $x = 2$ b) $x = 3$ c) $x = \pm 1$ d) $x = -2, x = -4$ e) $x = 0, x = \pm 3$ f) $x = 0, x = \pm 2$
 2a) $x = 5, y = -2$ b) $x = 0, y = x$ c) $x = \frac{1}{3}, y = -3$ d) $x = 1, y = x + 6$ e)
 f) $x = \pm \sqrt{\frac{5}{3}}, y = -2$
 3a) $\{x/x \in \mathbb{R}, x \neq 5\}$ b) none c) -5
 4a) $\{x/x \in \mathbb{R}, x \neq -5, x \neq 1\}$ b) $\forall x = -5 \ x = 1 \ \text{hay} = 0$ (c) $x = -1$
 5a) $\{x/x \in \mathbb{R}, x \neq 2\}$ b) $\forall x = 2 \ \text{oa} \ y = x + 1$ (c) $x = 3, x = -2$
 6a) i) 3875, ii) 3065.04 b) approaches a value of \$3000 c) no, VA at $y = 3000$

TRIGONOMETRIC FUNCTIONS

- Convert (a) to degrees $\frac{\pi}{5}$ (b) to radians 155°
- Determine the exact value of
 (a) $\cos \frac{3\pi}{4}$ (b) $\tan \frac{7\pi}{6}$ (c) $\sec \frac{5\pi}{3}$ (d) $\cot \frac{\pi}{4}$ (e) $\sin \frac{-5\pi}{6}$
- Determine the length of the arc given: angle measure 2.345 radians and radius 3cm
- Write the expression as a single real number.
 (a) $\sin \frac{\pi}{6} \cos \frac{\pi}{2} - \cos \frac{\pi}{6} \sin \frac{\pi}{2}$ (b) $\sin \frac{\pi}{6} \cos \frac{\pi}{3} + \cos \frac{\pi}{6} \sin \frac{\pi}{3}$
 (c) $\cos \frac{5\pi}{6} \cos \frac{2\pi}{3} + \sin \frac{5\pi}{6} \sin \frac{2\pi}{3}$
- Given $\cos x = \frac{-3}{5}$ $\pi \leq x \leq \frac{3\pi}{2}$, find $\cos 2x$ and $\sin 2x$. Which quadrant is $2x$ in?
- Given $\cos x = \frac{2}{5}$, $0 < x < \frac{\pi}{2}$ Determine the value of $\sin \frac{x}{2}$.
- Solve for x given (a) $5\cos x + 3\cos 2x = 3$, $0 \leq x \leq \frac{\pi}{2}$ Round to 4 decimal places.
 (b) $\cos^2 x - \sin^2 x = \frac{1}{2}$, $0 \leq x \leq 2\pi$ (c) $\sin 2x - \cos x = 0$ $-\pi \leq x \leq \pi$
- prove (a) $\frac{\sec x + 1}{\tan x} = \frac{\tan x}{\sec x - 1}$ (b) $\frac{\cos^4 x - \sin^4 x}{1 - \tan^4 x} = \cos^4 x$ (c) $\frac{\tan x - \sin x}{2 \tan x} = \sin^2 \left(\frac{x}{2} \right)$
- If $\sin x = \frac{-12}{13}$ with $\pi \leq x \leq \frac{3\pi}{2}$ and $\sec y = \frac{13}{12}$ with $\frac{3\pi}{2} \leq y \leq 2\pi$ Find $\cos(x + y)$
- Given $y = -2\sin\left(2\theta - \frac{\pi}{3}\right) + 3$ State (a) the amplitude (b) the length of the period
 (c) the phase shift (d) graph 2 cycles

11. Given $y = \tan(\theta - \frac{\pi}{2})$ (a) State the period (b) graph 3 cycles

12. Solve (a) $4\cos^2 x + 4\cos x + 1 = 0$ on $-2\pi \leq x \leq 2\pi$
 (b) $-10\cos^2 x - 3\sin x + 9 = 0$ on $-2\pi \leq x \leq 2\pi$
 (c) $2\tan^2 x + 5\tan x + 3 = 0$ on $[0, 2\pi)$

ANSWERS: TRIGONOMETRIC FUNCTIONS

1(a) 36° (b) $\frac{31\pi}{36}$ 2a) $\frac{-1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$ (c) 2 (d) -0.5 3) 7.035 4a) $\frac{-\sqrt{3}}{2}$

4(b) 1 4(c) $\frac{\sqrt{3}}{2}$

5. $\cos 2x = \frac{-7}{25}$ $\sin 2x = \frac{24}{25}$ second quadrant

6. $\frac{\sqrt{30}}{10}$

7 a) $x = 0.8411$ b) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ c) $x = \frac{-\pi}{2}, \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$ 9. -1

10a) 2 b) π c) $\frac{\pi}{6}$ right 11) π

12a) $\frac{-5\pi}{6}, \frac{-7\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}$ b) $\frac{-7\pi}{6}, \frac{-11\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$, 3.343, 6.0818, -0.2014, -2.9402 c) $\frac{3\pi}{4}, \frac{7\pi}{4}$, 2.5536, 5.6952

EXPONENTIAL AND LOGARITHMIC FUNCTIONS

1. Solve the equation $\log_2(x+1) = 3$

- (a) 5 (b) 6 (c) 7 (d) 8 (e) 9

2. Solve the following equation to 3 decimal places: $3^x = 10$

- (a) 1.820 (b) 1.922 (c) 1.971 (d) 2.034 (e) 2.096

3. The domain of $y = \log(9 - x^2)$, $x \in R$, is

- (a) $x > 0$ (b) $x > 3$ or $x < -3$ (c) $-3 \leq x \leq 3$
 (d) $x \geq 3$ or $x \leq -3$ (e) $-3 < x < 3$

4. A bacteria culture is growing according to the function $N(t) = 100(2)^t$, where $N(t)$ is the population after t hours. How long will it take the population to grow to 51200?

- (a) 8 h (b) 9 h (c) 10 h (d) 11 h (e) 12 h

5. Solve $\log_8 x + \log_8 (x-12) = 2$

ANSWERS: EXPONENTIAL AND LOGARITHMIC FUNCTIONS

1. c 2. e 3. e 4. b 5. $x = 16$, $x = -4$ extraneous

RATES OF CHANGE

1. In 2000 the population of the world was 6 million. The population growth of the world is growing exponentially and can be modeled by the equation:

$$p(t) = 6(2^{\frac{t}{13}}), \text{ where } p \text{ is the population in millions } t \text{ years after the year 2000.}$$

- a) Calculate the average population growth from the years 2003 to 2007 ($t=3$ to $t=7$)
 b) Approximate the instantaneous growth rate at the year 2007 ($t=7$)

ANSWERS: RATES OF CHANGE

- a) 418444 people per year b) 464651 people per year

COMBINATION OF FUNCTIONS

1. Given that $f(x) = x^3 - x^2 + 5x$ and $g(x) = x^3 + 5x^2 - 2$ determine $(g-f)(x)$.

- (a) $4x^2 + 5x - 2$ (b) $-6x^2 + 5x + 2$ (c) $6x^2 - 5x - 2$
 (d) $-6x^2 + 5x - 2$ (e) $6x^2 - 5x + 2$

2. Given that $f(x) = 3x^2$ and $g(x) = 2x + 1$ then $f(g(x))$ equals:

- (a) $6x^3 + 3x^2$ (b) $6x^2 + 1$ (c) $4x^2 + 4x + 1$
 (d) $3x^2 + 2x + 1$ (e) none of these

3. Using interval notation, the domain of f is $[-3, 5]$ and the domain of g is $[-2, 7]$. On what domain is $f+g$ defined?

- (a) $[-3, 7]$ (b) $[-2, 5]$ (c) $[-3, -2]$ (d) $[5, 7]$ (e) $[-2, 7]$

4. For a car traveling at a constant speed, the distance driven, d kilometers, is represented by $d(t) = 80t$, where t is the time in hours. The cost of gasoline, in dollars, for the drive is represented by $C(d) = 0.09d$. Evaluate $C(d(10))$.

- (a) 38 (b) 44 (c) 72 (d) 80 (e) 96

5. The speed of a car, v kilometers per hour, at a time of t hours is represented by $v(t) = 40 + 3t + t^2$. The rate of gasoline consumption of the car, c litres per kilometer at a speed of v kilometers per hour is represented by the equation below. Write $c(v(t))$, the rate of gasoline consumption as a function of time. Calculate $c(v(4))$ accurate to three decimals.

[3]

$$c(v) = \left(\frac{v}{500} - 0.1 \right)^2 + 0.15$$

ANSWERS: COMBINATIONS OF FUNCTIONS

1. c 2. e 3. b 4. c

$$5. \quad c[v(t)] = \left[\frac{40 + 3t + t^2}{500} - 0.1 \right]^2 + 0.15$$

$$c[v(4)] = 0.151 \text{ L/km}$$